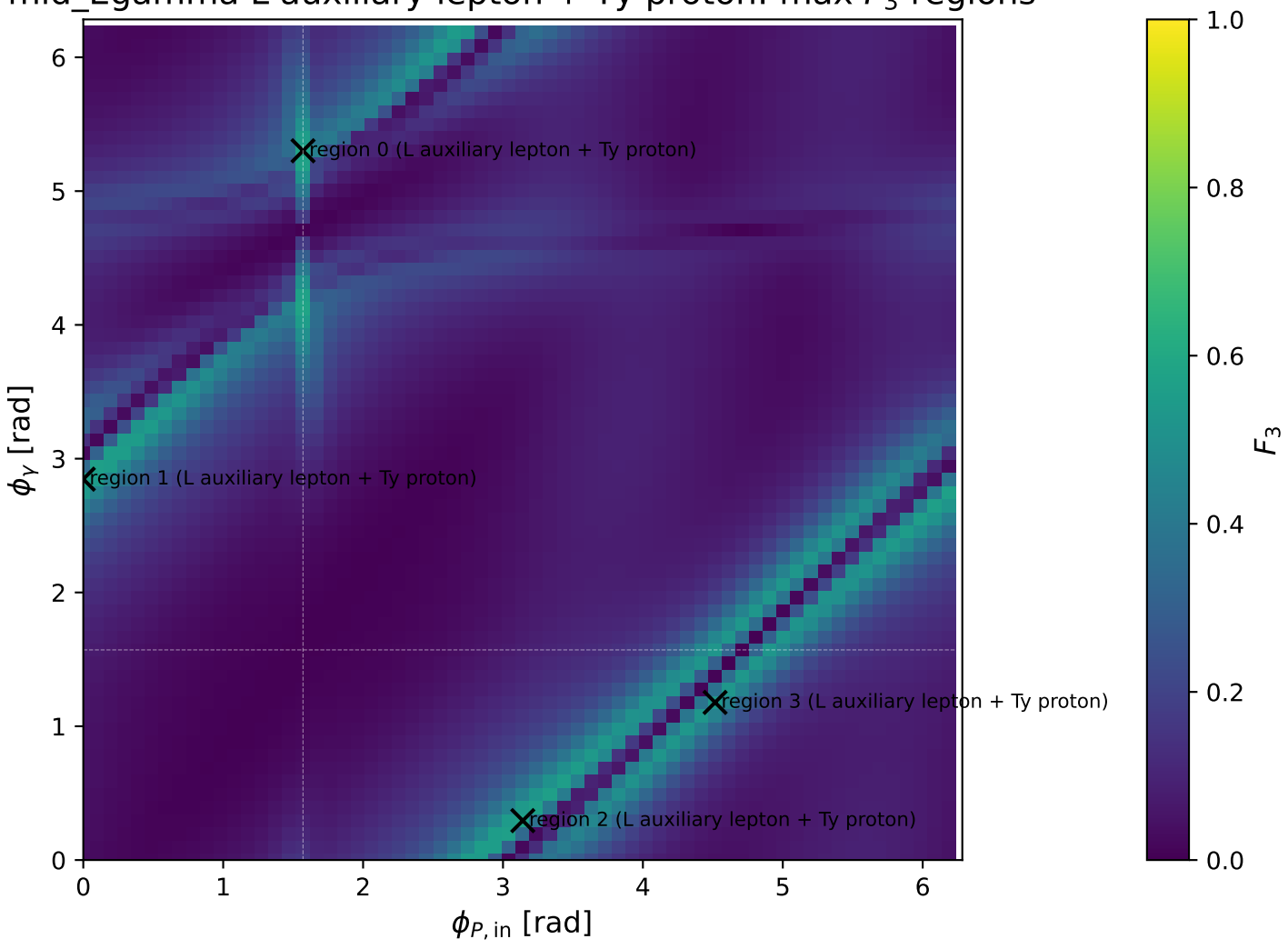
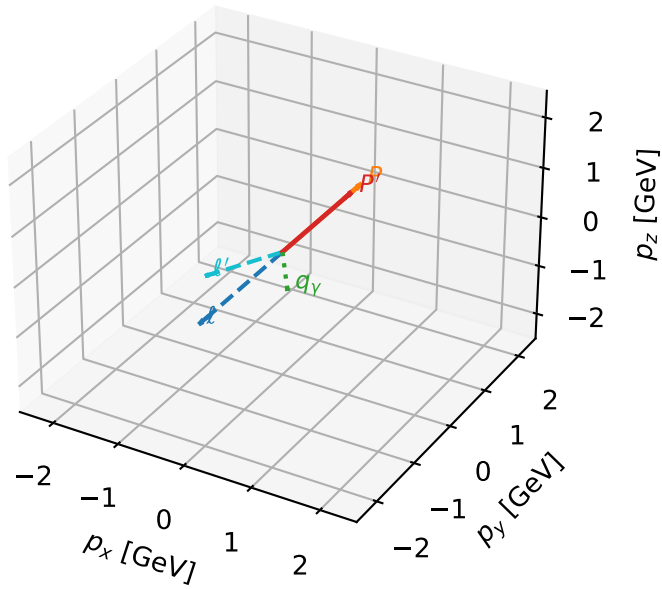


# mid\_Egamma L auxiliary lepton + Ty proton: max $F_3$ regions



# L auxiliary lepton + Ty proton characteristic max $F_3$ configuration

3D momenta

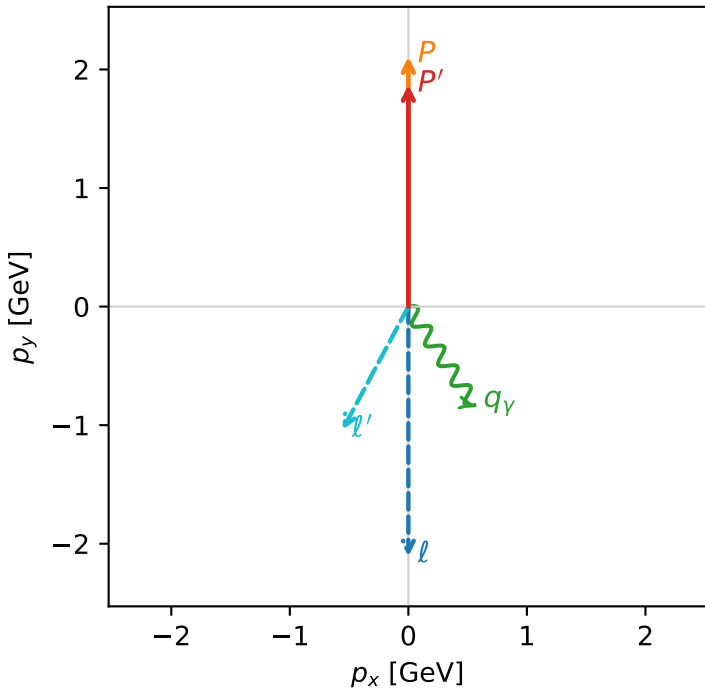


f3\_mid\_egamma\_lty\_region\_0 F\_3=0.583757  
 region: mid\_Egamma\_LTy\_region\_0  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_mid\_Egamma  
 incoming state:  $|L_l\rangle \otimes |T_{Y,p}\rangle$

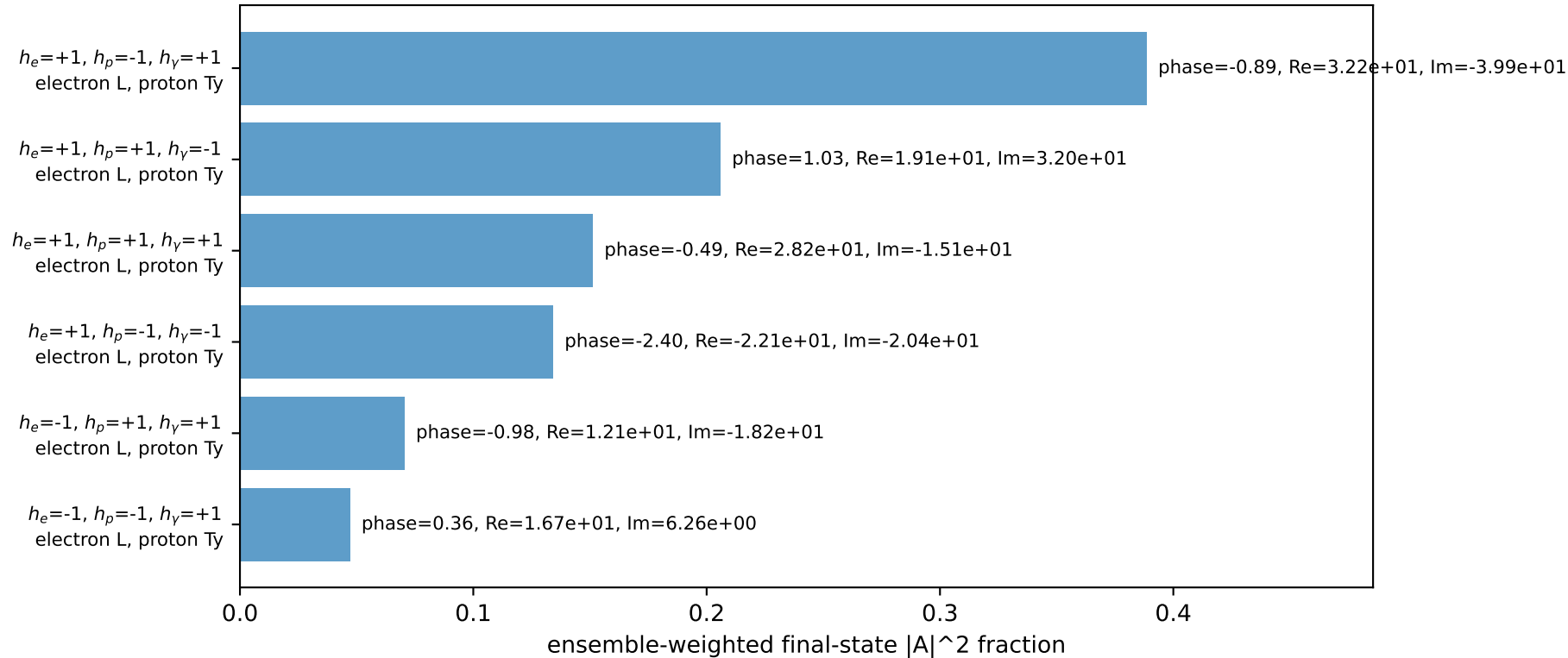
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.10694$ ,  $|\vec{P}'|=1.87089$   
 $\theta_{in}=1.57079$ ,  $\phi_l=4.71239$ ,  $\phi_p=1.5708$   
 $E_Y=1$ ,  $\phi_Y=5.30144$   
 $Q^2=0.829595$ ,  $x_B=0.102085$ ,  $t=-0.0101624$ ,  $W^2=8.17674$ ,  $y=0.413857$   
 $\theta(l', \gamma)=61.8745$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.3322$   $p_x= -0.0000$   $p_y= -2.1069$   $p_z= -0.0000$   
 $P$   $E= 2.3063$   $p_x= 0.0000$   $p_y= 2.1069$   $p_z= 0.0000$   
 $l'$   $E= 1.5457$   $p_x= -0.5556$   $p_y= -1.0394$   $p_z= 0.0000$   
 $P'$   $E= 2.0929$   $p_x= 0.0000$   $p_y= 1.8709$   $p_z= 0.0000$   
 $q_Y$   $E= 1.0000$   $p_x= 0.5556$   $p_y= -0.8315$   $p_z= 0.0000$

Transverse plane

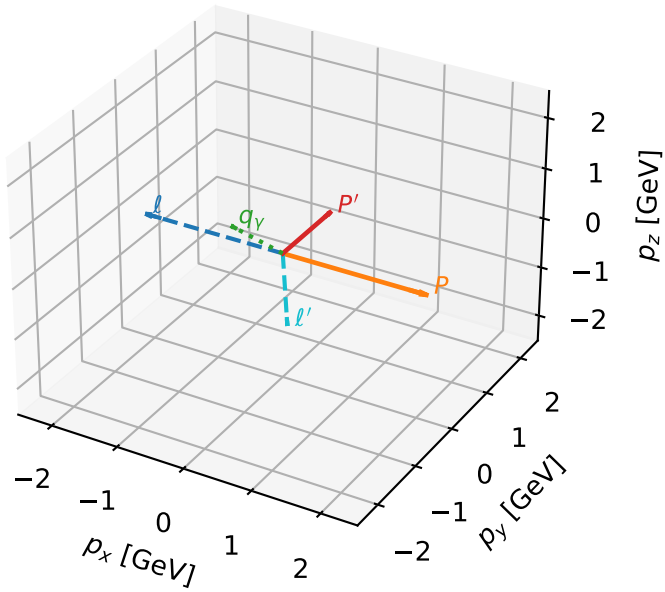


Leading initial-ensemble/final-state components (N=6, retained=99.7%)



# L auxiliary lepton + Ty proton characteristic max $F_3$ configuration

3D momenta

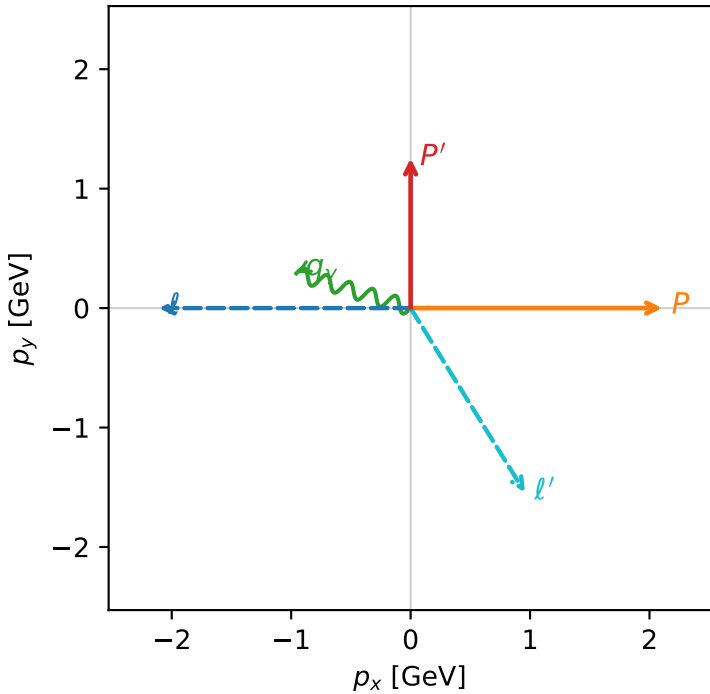


f3\_mid\_egamma\_lty\_region\_1  $F_3=0.565251$   
 region: mid\_Egamma\_LTy\_region\_1  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_mid\_Egamma  
 incoming state:  $|L_l\rangle \otimes |T_{y,p}\rangle$

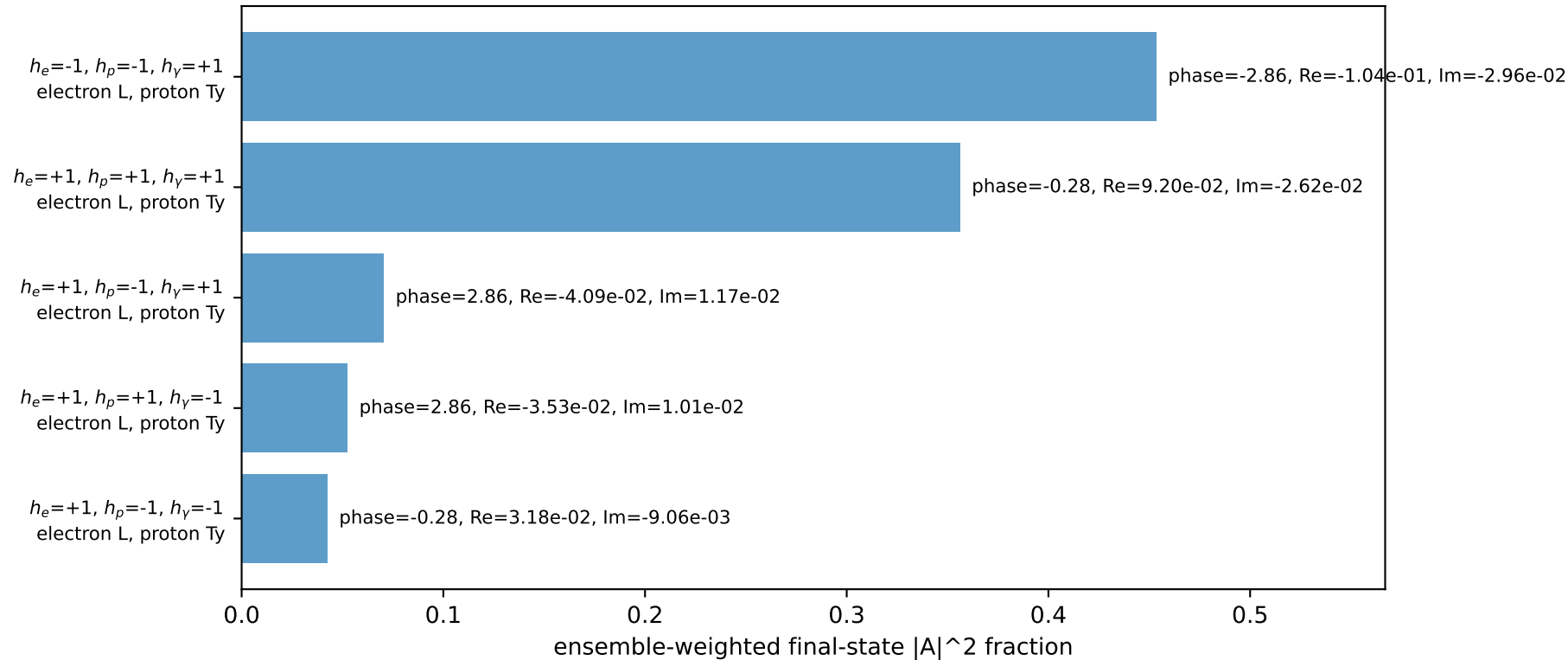
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.10694$ ,  $|\vec{P}'|=1.25318$   
 $\theta_{in}=1.57079$ ,  $\phi_l=3.14159$ ,  $\phi_P=0$   
 $E_Y=1$ ,  $\phi_Y=2.84707$   
 $Q^2=11.7026$ ,  $x_B=0.829634$ ,  $t=-5.46066$ ,  $W^2=3.28298$ ,  $y=0.71836$   
 $\theta(l', \gamma)=138.674$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.3322$   $p_x= -2.1069$   $p_y= -0.0000$   $p_z= -0.0000$   
 $P$   $E= 2.3063$   $p_x= 2.1069$   $p_y= 0.0000$   $p_z= 0.0000$   
 $l'$   $E= 2.0732$   $p_x= 0.9569$   $p_y= -1.5435$   $p_z= 0.0000$   
 $P'$   $E= 1.5653$   $p_x= 0.0000$   $p_y= 1.2532$   $p_z= 0.0000$   
 $q_Y$   $E= 1.0000$   $p_x= -0.9569$   $p_y= 0.2903$   $p_z= 0.0000$

Transverse plane

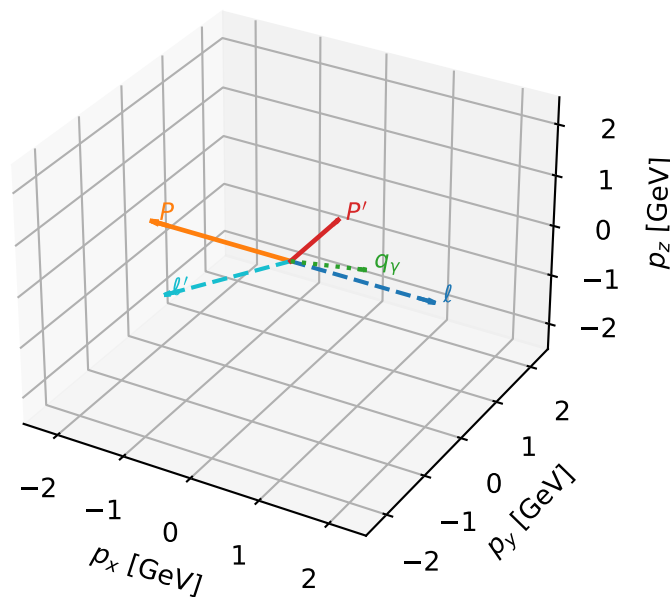


Leading initial-ensemble/final-state components (N=5, retained=97.5%)



L auxiliary lepton + Ty proton characteristic max  $F_3$  configuration

3D momenta

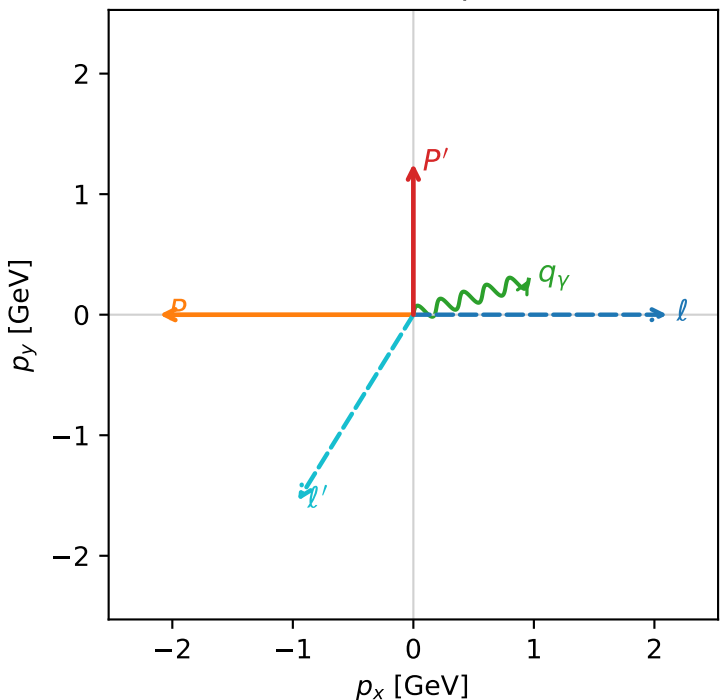


f3\_mid\_egamma\_lty\_region\_2 F\_3=0.565251  
region: mid\_Egamma\_LTy\_region\_2  
outgoing-state purity: 1  
kinematic point: high\_s\_high\_theta\_in\_mid\_Egamma  
incoming state:  $|L_l\rangle \otimes |T_{y,p}\rangle$

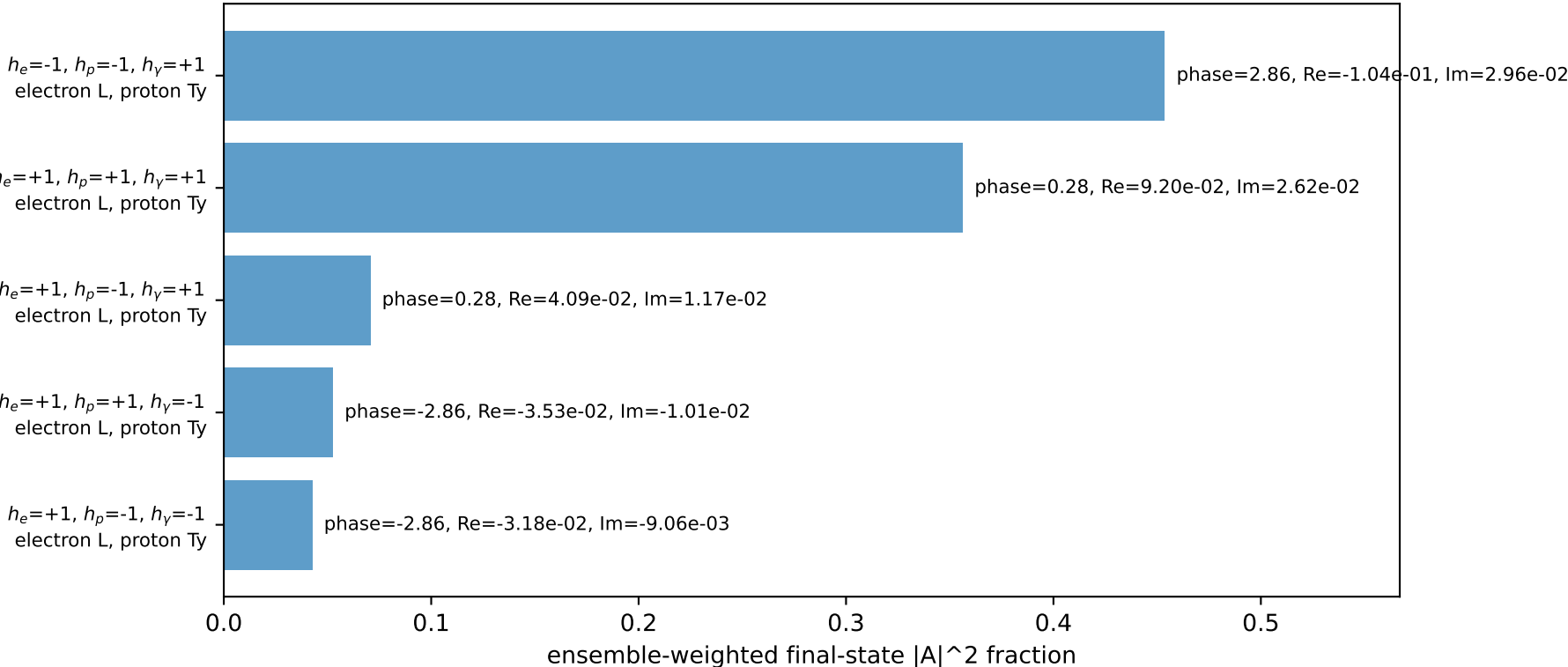
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.10694$ ,  $|\vec{P}'|=1.25318$   
 $\theta_{in}=1.57079$ ,  $\phi_l=0$ ,  $\phi_p=3.14159$   
 $E_\gamma=1$ ,  $\phi_\gamma=0.294524$   
 $Q^2=11.7026$ ,  $x_B=0.829634$ ,  $t=-5.46066$ ,  $W^2=3.28298$ ,  $y=0.71836$   
 $\theta(l', \gamma)=138.674$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.3322$   $p_x= 2.1069$   $p_y= -0.0000$   $p_z= -0.0000$   
 $P$   $E= 2.3063$   $p_x= -2.1069$   $p_y= 0.0000$   $p_z= 0.0000$   
 $l'$   $E= 2.0732$   $p_x= -0.9569$   $p_y= -1.5435$   $p_z= 0.0000$   
 $P'$   $E= 1.5653$   $p_x= 0.0000$   $p_y= 1.2532$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.0000$   $p_x= 0.9569$   $p_y= 0.2903$   $p_z= 0.0000$

Transverse plane

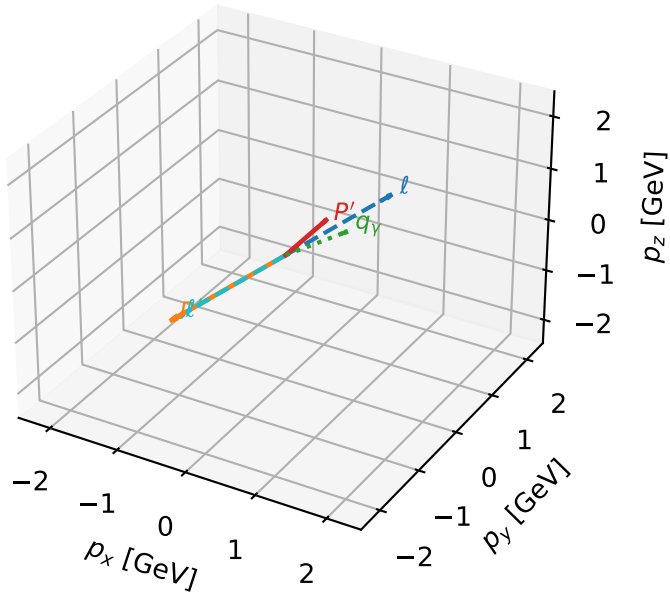


Leading initial-ensemble/final-state components (N=5, retained=97.5%)



# L auxiliary lepton + Ty proton characteristic max $F_3$ configuration

3D momenta

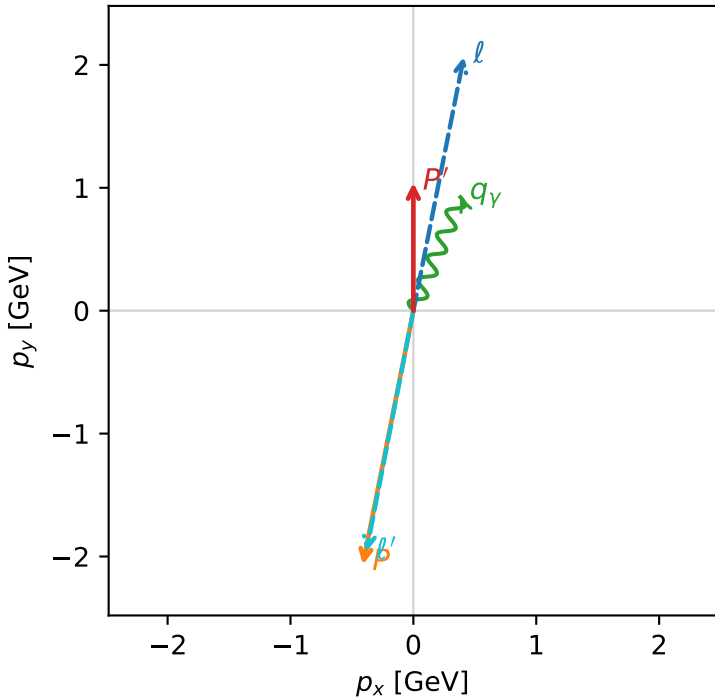


f3\_mid\_egamma\_lty\_region\_3 F\_3=0.529703  
 region: mid\_Egamma LTy\_region\_3  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_mid\_Egamma  
 incoming state:  $|L_\ell\rangle \otimes |T_{y,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.10694$ ,  $|\vec{P}'|=1.04074$   
 $\theta_{in}=1.57079$ ,  $\phi_l=1.37445$ ,  $\phi_p=4.51604$   
 $E_\gamma=1$ ,  $\phi_\gamma=1.1781$   
 $Q^2=16.8706$ ,  $x_B=0.950473$ ,  $t=-9.00419$ ,  $W^2=1.75894$ ,  $y=0.903938$   
 $\theta(l', \gamma)=168.522$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.3322$   $p_x= 0.4110$   $p_y= 2.0665$   $p_z= -0.0000$   
 $P$   $E= 2.3063$   $p_x= -0.4110$   $p_y= -2.0665$   $p_z= 0.0000$   
 $l'$   $E= 2.2375$   $p_x= -0.3827$   $p_y= -1.9646$   $p_z= 0.0000$   
 $P'$   $E= 1.4011$   $p_x= 0.0000$   $p_y= 1.0407$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.0000$   $p_x= 0.3827$   $p_y= 0.9239$   $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=3, retained=99.6%)

